

Research Article

Baccalaureate nursing education curricula in the People's Republic of China: Status, issues and reforms

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Abstract

This article examines baccalaureate nursing education curricula in the People's Republic of China. Three 5-year curricula and one 4-year curriculum were content analyzed and contrasted. Findings of this study suggested that: (i) the biomedical model dominated the three 5-year traditional curricula; (ii) political-ideological content permeated all four curricula; and (iii) the reformed curriculum at Peking Union Medical College was a genuine effort toward building a new nursing curriculum model that intended to differentiate nursing education from medical education. In addition, major themes of recent curriculum reforms were highlighted. Recommendations were suggested to improve the outcomes of future curriculum endeavors. Explicit as well as implicit comparisons with an ideal-typical American generic baccalaureate nursing curriculum were made when appropriate.

Key words

baccalaureate nursing education, China, curriculum, reform.

INTRODUCTION

China has a multi-tier nursing education system with programs at the secondary, associate, baccalaureate and graduate levels (Xu *et al.*, 2000). Although there are a handful of graduate programs currently in operation, baccalaureate nursing education is, to a large extent, perceived as the pinnacle of the system. Baccalaureate nursing education has been expanding rapidly since its restoration at Tianjin Medical College (now Tianjin Medical University) in 1983 after a 30-year abolition of collegiate nursing education in the country. The latest statistics indicate that baccalaureate nursing programs increased from 18 in 1998 (Chang, 1999) to 67 in 1999 (Li, 2001). However, the size and admission capacity of these programs remain limited. The proportion of baccalaureate graduates in the entire 1.2 million nurse population is less than 1% (Shen, 1998a).

Curriculum is at the crux of nursing education. While curriculum defies easy definition, it refers to, for the purpose of this study, a prescribed program of studies that defines course offerings, time units each course carries, and the suggested sequencing of courses. This study is an examination of the official curriculum or the 'legitimate curriculum' (Bevis, 1989a) of current baccalaureate nursing education in the People's Republic of China. In addition, explicit as well as implicit comparisons are made with an ideal-typical American generic baccalaureate nursing curriculum when appropriate.

Research on Chinese baccalaureate nursing education is extremely scarce in available databases in English such as CINAHL. Although there are a few published studies (DeSantis, 1988, 1995; Davis *et al.*, 1990; Davis *et al.*, 1992; Dai, 1990; Chan & Wong, 1999; Wong *et al.*, 2000), none of them treats the baccalaureate nursing education curriculum as their focus. Moreover, some of these studies are either based on secondary analysis or are non-data-based. The only research relevant to the present study is by French *et al.* (1996), which compares baccalaureate nursing degrees from Australia, China, Hong Kong, South Africa, United Kingdom and United States of

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America through examining their respective curricula to uncover similarities and differences. Some of the major findings were: (i) most curricula did not include anything about the humanities with few indicators valuing the artistic, humanistic, and caring aspects of nursing as well as professional issues; (ii) biological science was perceived as central to most courses, and nurses were taught alongside students from other disciplines in some cases; (iii) there were two ways to organize nursing studies: role-related approach emphasizing roles of the professional nurse and the medical model approach that defines nursing content along the lines of specialty areas in clinical medicine; and (iv) little clinical experience was provided outside hospital contexts.

METHODS

Content analysis was employed as the primary method of inquiry in this descriptive study. The curricula of all baccalaureate nursing programs in China were targeted for this study. The only predetermined criterion for inclusion was that the selected programs must have at least one class of graduates by the end of 2000 to ensure that the curricula had gone through one complete cycle. Requests for curricula and related documents were sent to all known baccalaureate nursing programs in China ($n=22$ at the time of mailing) in early 2000. The request and a follow up 2 months after the initial mailing generated curriculum data from four baccalaureate programs (18% return rate) that were believed to be representative of the hierarchic baccalaureate nursing system. The four programs were Peking Union Medical College (PUMC) representing the first-tier programs that served as national leaders in nursing education, including curriculum reforms; Shanghai Second Medical University (SSMU) representing the second-tier programs that had an established reputation, yet followed the lead of the first-tier programs; and Bengbu Medical College (BMC) and Changzhi Medical College (CMC) representing the third-tier programs that were relatively new and at the periphery of the baccalaureate nursing education system. PUMC had a 4-year curriculum while the remaining three programs retained the traditional 5-year curricula that shared the same program length with medical programs.

Two explanations were plausible for the less-than-satisfactory return rate: (i) many programs were undergoing major curriculum revisions without a finalized version at time of the request; and (ii) concern over competition and rivalry may have been a more important factor affecting the return rate. Although efforts were made to improve the sample

size and random sampling should have been employed ideally, these two issues potentially affecting the validity and generality of the study were of minor concern after all, primarily due to two reasons. First, the number of existing baccalaureate nursing programs in China was very small (22 programs known at the time of the study). Second, China is a highly centralized country in terms of educational administration and management, including curriculum approval, implementation and revision. The state has 'a hegemonic role in defining school curricula through its bureaucratic, centralized, and coercive mechanisms' (Adamson & Morrison, 1997) although the state-controlled process has evolved into a more complex mechanism that displays an increased degree of pluralism in response to social and educational changes. This means that, except in specially approved cases, the essential curriculum content is dictated by government regulations and is unlikely to vary significantly between programs, as evidenced by the available curriculum data presented later.

Content analysis was conducted with the four returned curricula with descriptive statistical analysis. This analytic process consisted of two steps: (i) the three traditional 5-year curricula at BMC, CMC, and SSMU were compared; and (ii) the three traditional five-year curricula as a whole were contrasted with the reformed PUMC 4-year curriculum. To maintain validity, a course classification scheme, which is explained in detail in the next section, was developed with operational definitions for each course category prior to the coding process. The data coding and analysis procedures were as follows. First, two team researchers independently categorized all of the courses in the four curricula into a Microsoft Excel database, employing the course classification scheme. Second, to minimize subjectivity, coding results were compared and differences resolved through group consensus, or by consulting with two pre-identified published experts when a course classification issue went beyond the collective expertise of the team. Both scholars had extensive knowledge of, and wide-ranging experience with, the Chinese health-care system, as well as medical and nursing education in China. Third, descriptive statistics were computed and analyzed.

Additionally, the content analysis of the curriculum data was complemented by a systematic review of studies on Chinese baccalaureate nursing curriculum published between 1990 and 2000 in both the English and the Chinese languages. These studies not only augmented the available curriculum data, but also provided insights for data interpretation in the present study by keeping the findings in perspective.

One logistic issue needs to be addressed. The exact amount of time assigned for practicum in the three 5-year curricula was either unstated or unspecified although two curricula did indicate a 48-week practicum in the fifth academic year as a general statement. In order to make comparisons across all three curricula possible, a 48-week practicum was assigned to each of the three 5-year curricula with the assumption that student nurses provided supervised care 8 hours a day, 5 days a week.

PROFILE OF BACCALAUREATE NURSING EDUCATION CURRICULA

Curricula configuration and findings

From the vantage point of curriculum structure, Chinese baccalaureate nursing programs are divided into two basic categories: 4-year programs and 5-year programs, with the latter dominating the system. In fact, PUMC was the only 4-year program in the entire country at the time of data collection for this study (Anli Jiang, pers. comm., 2001). Required courses and electives are the two basic categories of courses in the curriculum with subcategories within both. Subcategories under required courses include general education courses, professional foundation courses, and nursing courses (including practicum). Subcategories under electives are restrictive electives and free electives.

General education courses refer to the category of course offerings that are intended to lay a common foundation for baccalaureate graduates regardless of academic majors. They are the shared core in a college or university curriculum such as Marxist Philosophy, English, Advanced Mathematics, Biology, Chemistry and Computing Basics. Professional foundation courses are the category of courses that serve as the foundation for professional nursing studies such as Anatomy, Biochemistry, Physiology, Pharmacology, Bio-statistics, and Epidemiology. Nursing

courses refer to professional studies that address the practise, theory, research, education, services, and the profession in nursing. Practicum is an integral part of professional nursing studies. It is defined as full-time clinical experience under the supervision of a designated preceptor-mentor.

Electives refer to a group of courses that are intended to enlarge and/or supplement the existing instructional offerings. There are two subcategories within elective courses: restrictive and free. Restrictive electives refer to choices among a designated group of courses, usually related to one's major. In contrast, free electives are those courses that one can liberally choose among the entire college–university instructional offerings that might or might not be related to one's major.

In addition to formal courses, other requirements that were typically included in an entire prescribed curriculum were admission orientation (1 week), examinations (15–16 weeks), military training and productive manual labor (2–4 weeks), community services (2 weeks), graduation and job assignment related activities (2 weeks), and summer and winter breaks (36–37 weeks). SSMU also required a graduation thesis for which 8 weeks were assigned for its preparation and two additional weeks for its defense. Due to their non-instructional nature and/or unavailability of comparable data, these activities have been excluded from examination and analysis.

General education courses

There were marked similarities in both the course offerings and the allocated time to each course in this course category across the three traditional curricula. This was no surprise considering that China is a highly centralized country. A number of characteristics stood out readily from an immediate observation and comparison of the three curricula (Table 1). First, the dominant feature of the course listing was the pervasiveness of political-ideological content with an

Table 1. Time allocation by course category in three traditional 5-year curricula (hours)*

Course category	BMC (%)	SSMU (%)	CMC (%)	Mean (%)
General education	1091 (20.8)	1215 (22.7)	1248 (22.5)	1185 (22.0)
Professional foundation	1033 (19.7)	1180 (22.0)	1352 (24.5)	1188 (22.1)
Nursing	3121 (59.5)	2958 (55.3)	2928 (53.0)	3002 (55.9)
Total hours	5245 (100)	5353 (100)	5528 (100)	5375 (100)

* Electives not included due to unavailability of complete and comparable data. BMC, Bengbu Medical College; SSMU, Shanghai Second Medical University; CMC, Changzhi Medical College.

Table 2. Time allocation of general education courses in three traditional 5-year curricula (hours)

Course offerings	BMC	SSMU	CMC	Mean (mean % of general education)
Marxist Political Economy	40	40	40	40 (3.32)
Mao Zedong Thought	36	36	36	36 (2.99)
Marxist Philosophy	54	54	54	54 (4.49)
Deng Xiaoping Thought	70	70	72	71 (5.90)
Moral Education	—	51	50	51 (4.24)
Legal Study	30	34	34	33 (2.74)
Physical Education	142	144	142	143 (11.88)
English	355	396	320	357 (29.65)
Advanced Mathematics	34	36	50	40 (3.32)
Biology	42	42	50	45 (3.74)
Physics	72	72	90	78 (6.48)
Chemistry	135	160	190	162 (13.46)
Computing Basics	81	80	120	94 (7.81)
Total hours	1091	1215	1248	1185
Proportion of entire curriculum	20.8	22.7	22.6	22.0

BMC, Bengbu Medical College; SSMU, Shanghai Second Medical University; CMC, Changzhi Medical College.

average of 20.94% time allocation within general education, which translated into 4.69% of the total mean curriculum time. Nursing students, like their counterparts in other majors, were taught to ‘cherish the motherland, the people, and support the Chinese Communist Party’, to study and work diligently, to observe school disciplines and public order, and to respect their teachers.

Second, the status of English in the curricula was outstanding, with a mean of 29.65% share of total general education time devoted to this subject alone (Table 2). This finding reflected the government position that English was a vital tool for technology advancement and knowledge transfer, especially in the context of increasing globalization.

Third, the emphasis on the sciences was evident with a mean of 27% share within general education course category allocated to this cluster of subjects (Table 2). The total hours in general education courses as a percentage of total curricula across the three traditional programs ranged from 20.8 to 22.7 with a mean of 22.0 (Table 2).

Professional foundation courses

The three traditional curricula shared 11 (50%) courses in this subcategory. The proportion of professional foundation courses in the total curriculum time ranged from 19.7% to 24.46% with a mean of 22.10% (Table 3). Overwhelming evidence from the sampled programs supported the presumption that Chinese baccalaureate nursing education curriculum

was heavily influenced by the biomedical model. First, the entire course offerings mirrored the knowledge content for clinical medicine rather than nursing, with emphasis on the etiology, pathology, diagnosis, and treatment of disease processes. Second, nursing and medicine shared considerable content in this category. Many of the courses still retained the prefix ‘medical’ in their titles, indicating that those courses were originally developed for medical majors. These courses definitely belonged to the content domains of medical education. Later, when the baccalaureate nursing programs were established, nursing majors studied the same prerequisites alongside medical students. This arrangement was a plausible and realistic curriculum measure that would save valuable time for the nursing faculty to concentrate on the development of nursing courses. This argument is supported by the fact that medical students and baccalaureate nursing students took the same courses during the first 2 years of their studies (Xu *et al.*, 2000). Another interesting observation was that the extent of course offerings in this category appeared much more extensive and diverse than a typical American generic baccalaureate nursing curriculum.

Nursing courses

The share of nursing courses in the three traditional curricula ranged from 52.97% to 59.50% with a mean of 55.91% (Table 4). Several features were apparent. First, the influence of medical education was evident. The nursing specialty courses were arranged accord-

Table 3. Time allocation of professional foundation courses in three traditional 5-year curricula (hours)

Course title	BMC	SSMU	CMC	Mean
Anatomy	102	162	190	151
Histology & Embryology	74	90	90	85
Biochemistry	90	126	140	119
Physiology	72	126	140	113
Medical Microbiology	72	64	80	72
Medical Immunology	36	30	54	40
Pathophysiology	54	66	70	63
Laboratory Pathology	74	—	—	—
Parasitology	72	72	130	91
Pharmacology	36	44	50	43
Diagnostic Studies	72	90	108	90
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General Surgery	36	—	—	—
Imaging Diagnostics	45	44	—	45
Bio-statistics	72	36	—	54
Epidemiology	36	18	20	25
Medical Genetics	36	—	—	—
Preventive Medicine	—	36	110	73
Nursing Informatics	—	20	—	—
Managerial Psychology	—	18	—	—
Medical Aesthetics	—	18	—	—
Radiology	—	—	30	—
Total hours	1033	1180	1352	
Proportion of entire curriculum	19.70	22.04	24.46	22.07

BMC, Bengbu Medical College; SSMU, Shanghai Second Medical University; CMC, Changzhi Medical College.

ing to the subspecialties in clinical medicine. More specifically, the curriculum configuration was a copy of the medical education model. In addition, many of the course titles remained the same, only suffixed with 'nursing'.

Second, the offerings of Traditional Chinese Medicine Nursing and Health Education/General Pedagogy were unique in the Chinese baccalaureate nursing curriculum. Although the former was generally an introductory course that addressed the holistic and dialectic principles of Traditional Chinese Medicine with its derived knowledge and skills, the latter focused on the teaching theories and methods within both the clinical and the academic settings.

Third, compared with a typical American generic baccalaureate nursing curriculum, a few content areas were missing: community health; nursing research;

Table 4. Time allocation of nursing courses in three traditional 5-year curricula (hours)

Course title	BMC	SSMU	CMC	Mean
Introduction to Nursing	54	—	30	42
Health Education/General Pedagogy	54	36	—	45
Fundamentals of Nursing	205	162	222	196
Nursing Administration & Management	46	30	30	35
Nutrition	72	36	—	54
Internal Medicine Nursing	139	180	188	169
Surgical Nursing	157	180	220	186
Nursing/Medical Psychology	36	36	30	34
Obstetrical & Gynecologic Nursing	75	72	90	79
Pediatric Nursing	75	72	90	79
Psychiatric Nursing	42	36	22	33
Infectious Disease Nursing	41	—	—	—
Critical Care	36	—	—	—
Ophthalmologic Nursing	38	18	—	28
ENT Nursing	38	18	—	28
Oral Nursing	25	18	—	22
Dermatological Nursing	38	—	—	—
Nursing Ethics	30	30	—	15
Traditional Chinese Medicine & Nursing	—	36	—	—
Rehabilitative Nursing	—	24	—	—
Emergency Nursing	—	36	46	41
Gerontological Nursing	—	18	—	—
Epidemiological Nursing	—	—	40	—
Practicum (8 h per week × 48 weeks)	1920	1920	1920	1920
Total hours	3121	2958	2938	
Proportion of entire curriculum	59.50	55.26	52.97	55.91

BMC, Bengbu Medical College; SSMU, Shanghai Second Medical University; CMC, Changzhi Medical College.

management and leadership; professional issues; and role development.

Fourth, practicum was a major component in nursing courses that constituted a range of 61.52% to 65.35% of the time allocation within the category, or a range of 34.73% to 36.61% of the entire curricula time. It was a full-time clinical experience primarily in the format of a preceptorship under the direct supervision of a seasoned professional nurse designated as preceptor-mentor. It was the equivalent to the senior practicum in the American generic baccalaureate nursing curriculum in format and nature; however, it

ran throughout the entire fifth year in the three traditional curricula. Besides the practicum, little clinical experience beyond learning labs and clinical observation was provided across the curricula.

Electives

Incomplete information and wide variations with regard to electives made comparisons across the three traditional curricula impossible. While CMC listed 12 courses in restrictive electives ranging from 20 h to 46 h each, its requirement for non-restrictive electives was not stated nor specified. In contrast, BMC provided only 5 courses in this entire course category ranging from 18 h to 72 h each while SSMU merely stated that 30 credit hours, among which 10 must be in the humanities and social sciences, were required to graduate. An initial assessment based on examining the available data was that electives appeared to be a curriculum component with minimal importance. In fact, electives assumed such an obscure status in the curricula that their total hours were included in none of the curriculum grand totals. In addition, the number of electives was very limited, which, in turn, suggested that the freedom for course election was minimal.

Curriculum reform: PUMC School of Nursing

Peking Union Medical College School of Nursing has been a pioneer and forerunner in Chinese nursing education. It was the first baccalaureate nursing program in China, initially established in 1920. PUMC remained the only post-secondary nursing program in the country until 1952. It was the Soviet influence, but more importantly, the ideological conflict between China and the United States that lead to its eventual closure (Xu *et al.*, 2000). In the post-Mao era, PUMC was one of the first higher education institutions to restore the baccalaureate nursing program in the 1980s. It was the first to establish a collegiate school of nursing in 1996. At present, PUMC School of Nursing offers programs at the associate, baccalaureate and master's levels. In addition, it has served as one of the national centers for faculty training in nursing. Due to its long-standing leadership in nursing education, its geographic location, and its extensive network with international health-care organizations and institutions, PUMC has maintained its leadership in nursing education in China. It was designated jointly by the Ministry of Health and then State Education Commission as the reform trial site during 1995–2000 to implement a new baccalaureate nursing curriculum (Shen, 1998a, 1999, 2000).

Compared with the three traditional curricula discussed earlier, the PUMC curriculum was different in a number of ways (Table 5). First, the PUMC baccalaureate nursing curriculum was 4 years in length rather than 5 years. Based on its research (Shen, 1992; Shen *et al.*, 1994), PUMC decided to reduce the program length from 5 years to 4 years in 1996 in order to offer a curriculum that was designed to meet nursing's needs and objectives. The impetus for shortening the program length was to make it globally compatible. More importantly, the goal was to build a new curriculum model independent of the biomedical model to promote nursing's disciplinary development. Another goal was to curb the 'brain-drain' of baccalaureate nursing graduates into medicine by establishing a parallel curriculum with medical education (Shen, 1998a, 2000).

Second, the PUMC curriculum reduced the time allocation for professional foundation courses significantly. In the new curriculum the share of professional foundation courses in the total curriculum was 13.41% with a mean of 22.10%, considerably lower than its equivalents in the other three traditional curricula. The principal reason for the reduction was to increase nursing content both in depth and breadth to tackle the issue of nursing being replaced by medicine. One way to achieve the reduction was through rearrangement and integration of content areas from a nursing perspective rather than a medical perspective.

Third, integration and synthesis of nursing content were the most creative component of the curriculum and generated a curriculum model that drew a distinctive contrast with the three traditional curricula and medical curriculum. Instead of structuring nursing content according to body systems or life stages, nursing courses were conceptualized, configured, and integrated according to six broad categories of human needs and functions: social interaction, reproduction, nutrition–elimination, oxygenation, activity–rest, and cognition–perception (Shen, 1998a, 2000). Body systems and life stages have been the dominant organizing frameworks in the medical curriculum as well as in the nursing curriculum not only in China but also in other countries.

Fourth, an integrated approach was adopted with regard to clinical experience. It began once students started their professional nursing courses. This measure was targeted at bridging the gap between theory and practise in the traditional nursing curricula. In addition, the integration of didactic and experiential learning was consistent with learning principles.

Finally, the PUMC curriculum incorporated new content areas into the professional component such as

Table 5. Peking Union Medical College 4-year baccalaureate nursing curriculum

Course category/title	Hours (% of total)
General education	
Marxist Political Economy	40
Mao Zedong Thought	36
Marxist Philosophy	54
Deng Xiaoping Thought	70
Legal Studies	50
Physical Education	144
English	288
Human Biology I	132
Informatics	36
Moral Education	51
Inquiring and Logical Reasoning	36
State Affairs & Policy	54
Cultural Anthropology & Philosophy	40
Subtotal	1031 (23.51)
Professional foundation courses	
Biochemistry	54
Pharmacotherapeutics	72
Data Management	48
Managerial Psychology/Psychology	54
Health Care Delivery System	24
Communication	42
Human Biology II	90
Human Biology III	108
Growth & Development	54
Human-Environment Interaction	42
Subtotal	588 (13.41)
Nursing courses	
Nursing Education	54
Nursing Education Teaching Practicum	36
Fundamentals of Nursing	144
Nursing Management and Leadership	54
Nutrition	36
Bioethics & Professional Regulation	36
Traditional Chinese Medicine & Nursing	54
Health Assessment	108
Nursing Therapeutics	
Social Interaction (I)	108
Reproduction (II)	288
Nutrition-Elimination (III)	216
Oxygenation (IV)	216
Activity-Rest (V)	162
Cognition-Perception (VI)	216
Nursing Research	36
Research Project I	36
Research Project II	144
Practicum	480
Professional Development	36
Nursing Seminar/Special Topics	36
Community Health Nursing	270
Subtotal	2766 (63.08)
Grand Total	4385 (100)

community health, professional development, nursing education, and nursing research. The inclusion of a separate course on nursing education in the curriculum with a teaching practicum additionally clearly reflected PUMC's effort to promote the educator role both in the academic and the clinical settings to meet the state-defined expectations for its graduates. Further, PUMC emphasized nursing research with unprecedented weight. In addition to an independent course on nursing research, two research projects were in place, requiring students to complete a research paper before graduation (Shen, 1999). Such emphasis was designed as a curriculum measure to reduce and eventually eliminate the discrepancy between Chinese and international (American in particular) baccalaureate nursing education with regard to research training. In contrast, preparation for the researcher role was not evident from the course offerings in the three traditional curricula.

The ultimate goal of the new PUMC curriculum was to construct a new curriculum model for baccalaureate nursing education in China. This innovative curriculum claimed that it differentiated itself not only from the traditional nursing curricula but also from the medical education model and the popular body-system and life-cycle models in American baccalaureate nursing education (Shen, 2000). No evaluative data from consumers were available yet on the first PUMC baccalaureate class that graduated in 2000 after implementation of the new curriculum. However, the initial employers' evaluation of its first class of associate degree graduates in 1999 after the implementation of a reformed curriculum along similar principles was overwhelmingly positive (Shen, 2000).

DISCUSSION

Curriculum is the core of nursing education. Consistent data from the four scrutinized programs revealed that Chinese baccalaureate nursing curricula retained a strong commitment to political-ideological education at the turn of a new millennium. The four cardinal principles—keeping to the socialist road, upholding the people's democratic dictatorship, leadership by the Chinese Communist Party, and guidance by Marxist-Leninist-Maoist thought—were firmly observed in the process of developing curricula as well as in teaching and learning. According to Hayhoe (1989), History of the Chinese Communist Party, Political Economy, and Marxist-Leninist-Maoist Philosophy were three standard courses at the core of Chinese higher education over the past five decades,

taking up 10–15% of total curriculum time regardless of majors.

To a large extent, the contrast of the three traditional curricula has validated the findings of a multi-country curriculum study by French *et al.* (1996). To a considerable degree, the three traditional five-year curricula were water-downed versions of the medical education curriculum. A more critical stance in the literature denounced the situation as nursing being replaced with medicine (Hu & Yang, 1996; Shen, 1998a; You & Wang, 1999). The principal factor leading to such a precarious situation was that the development of nursing as an academic discipline was disrupted and halted for 30 years in China. Consequently, the field had to restart from scratch in the early 1980s, and thus relied heavily on medical education and medical faculty for curriculum development.

The influence of medical education was also rooted in the unique Chinese phenomenon that a significant proportion of the senior nursing faculty were initially trained as medical doctors. Due to a national nursing faculty shortage after the surge of baccalaureate nursing programs from the mid-1980s, these physicians were assigned and retooled to teach nursing. Despite the passionate efforts on the part of many of those former physicians to pursue nursing knowledge and skills, their teaching tended to focus on cure rather on care due to their original disciplinary training and the lack of formal nursing education.

Bevis (1989b), based on her international experience, criticized the dominance of the biomedical model in nursing education in some countries pointedly:

... in places where medical influence is predominant in nursing curriculum structure, the prerequisites are laden with the biological sciences far beyond the normal needs of first-level professional nursing and are lacking in the psychosocial and humanities base of nursing practise. It has been traditional in collegiate nursing programs to concentrate on the sciences as prerequisites to the exclusion of all but a few of the humanities courses. (p. 141)

Although the reformed PUMC curriculum may indicate the new thinking and directions in the ongoing national debate on nursing curriculum reform, it remains controversial. To some Chinese nurse educators, the curriculum restructuring approach is too radical, the curriculum too difficult to operate and, therefore, impractical; to others, the curriculum is a modified version of the American model and its applicability is questioned. It remains to be seen if the

PUMC reformed curriculum will find nation-wide acceptance. However, the reformed PUMC curriculum could have national implications for baccalaureate nursing education financing if data-based research proves that the PUMC curriculum retains quality. In light of the rapidly growing number of baccalaureate nursing programs, the national debate on the optimal program length for baccalaureate nursing will take on an added urgency.

The curriculum reform efforts represented by PUMC suggested that Chinese nurse educators were cognizant of both the need and challenge in developing a new nursing education model. Under the prevailing theme of curriculum reforms, three subthemes emerged from the reviewed literature.

From biomedical model to nursing-oriented curriculum model

Historically, the biomedical paradigm and medical education model influenced nursing education worldwide. This has been more so in China. Traditionally, the trilogy of medicine, nursing and pharmacotherapeutics in traditional Chinese medicine were interrelated and inseparable; instructionally, a considerable share of specialty nursing courses in baccalaureate nursing programs were taught by practising physicians due to the shortage of qualified nurse educators in these areas (Xu *et al.*, 2000); and administratively, nursing departments, and more recently, collegiate schools of nursing, were placed under medical colleges or universities. Consequently, the three baccalaureate nursing curricula mirrored the medical curriculum and reflected the cure-over-care medical education philosophy.

Bevis (1989b) summarizes the situation to the point:

In countries where nursing is still more closely tied to the traditions of medicine, conditions may exist where medical curriculum and physicians' perceptions of nursing content may greatly influence curriculum structure... In such settings, nursing content tends to be seen as deriving from medical content and appears as watered-down 'doctoring' instead of nursing. (p. 142)

China was not alone in that medicine made an imprinted mark on nursing and nursing education. As Bevis and Watson (1989) indicated, the first real departure from the medical model to a nursing one in American nursing occurred only 40 years ago when Abdullah, Martin, Beland and Matheny published *Patient-Centered Approaches to Nursing Care*

(Abdullah *et al.*, 1960). While empirical evidence indicated that the American nursing curriculum completed this paradigm shift by the early 1980s (DeBack, 1982), Chinese baccalaureate nursing curriculum is currently undergoing such fundamental changes.

From a specialized to a general curriculum model

Indigenous nurse educators perceived that the greatest deficit of the Chinese nursing education curriculum in general was the minimal presence of the humanities and social sciences (Wu & Jin, 1995; Hu & Yang, 1996; Kamata & Chen, 1997; Wang *et al.*, 1999). With the reforms in recent years, the curriculum landscape has changed considerably. Categorically, more courses in humanities and social sciences have been incorporated to reflect the paradigm shift in nursing education (You & Wang, 1999). However, the Chinese nursing curriculum as a whole was still perceived predominantly as 'physiologically based and disease-oriented' (Chan & Wong, 1999). The current study has confirmed the validity of this perception with the three traditional curricula.

Based on her international experience, Bevis (1989b) rationalized the incorporation of more humanistic content into the nursing curriculum while forcefully criticizing the dominance of the biomedical model in nursing education:

Given that nursing is as much an interactive discipline as a scientific one, and given that it is developing as a human science rather than continuing in the traditional science posture it once pursued, the humanities should occupy as strong a place as the sciences. It is in humanities that one finds the vicarious paradigm experiences that teach us about beauty, suffering, violence, peace, and human connectedness. (p. 141)

In a more fundamental sense, the reform was a direct result of the debates and comparative study on the Soviet and American baccalaureate education philosophies. While the Soviet model favored in-depth specialized knowledge compartmentalized into narrowly defined fields by occupation, it traded off knowledge breadth. One of the downsides of the Soviet model was fragmented knowledge without a broad foundation for future development. In contrast, the American model represented the other end of the spectrum; that is, relatively speaking, more breadth and less depth. Before 1949, Chinese higher education was modelled on American colleges and universities. After 1949, it was redesigned after the Soviet model, which has been challenged after China

implemented its open-door policy since the late 1970s (Hayhoe, 1989). Since then, there has been increasing pressure to find an eclectic approach to Chinese higher education.

Clinical instruction: Bridging the gap between theory and practise

Clinical teaching and learning are an integral part of nursing education as some nursing knowledge and many nursing skills can only be learned, applied and mastered through clinical experience. More importantly, clinical experience is critical in transforming a student nurse into a professional nurse. A major theme of clinical instruction reform is to bridge the divide between theory and practise. A national consensus to engage students in clinical instruction as early as possible is emerging (Kamata & Chen, 1997; Shen, 1998a, 2000; You & Wang, 1999). Such a measure is considered a move toward the international norm. Pedagogically, such a clinical arrangement is more conducive to learning and therefore should be more effective and efficient with regard to learning outcomes. However, empirical studies are needed to test this theory-based hypothesis.

CONCLUSIONS

Chinese baccalaureate nursing education curricula are undergoing significant changes, both in quantitative and qualitative terms. While these curriculum reform efforts demonstrated the vitality of nursing as a growing academic field, the systematic literature review conducted for this study revealed that most reform initiatives were piece-meal and fragmented with minimal conceptual guidance and lacked data-based research for support and legitimacy. Therefore, future curriculum reforms should adopt an integrated approach. Moreover, curriculum reform initiatives should become theory-driven or guided by conceptual frameworks. An encouraging indicator was that Chinese nurse educators are becoming increasingly cognizant of the need and imperative to conduct empirical research with scientific rigor to support curriculum reform endeavors (Yang *et al.*, 1997; Shen, 1998b, 1999).

Comparative studies on curricula are in high demand in current nursing education research, especially in the context of globalization. Philosophically, comparative studies present a mirror through which to examine oneself. Educationally, comparative studies provide the foundation for transnational credit transfer and student exchange initiatives. However, case (country) studies are the precursors for

comparative research. It is hoped that this study will stimulate further interest in conducting more comparative studies to identify the universality as well as the diversity in global nursing education curriculum.

Limitation

One weakness with the procedure in this study was the determination of course content solely based on their titles. The potential flaw of this face-value approach was that it may have missed or added some content areas that should have been included or excluded in data analysis. Ideally, future studies should examine the syllabus of each course to determine its content.

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